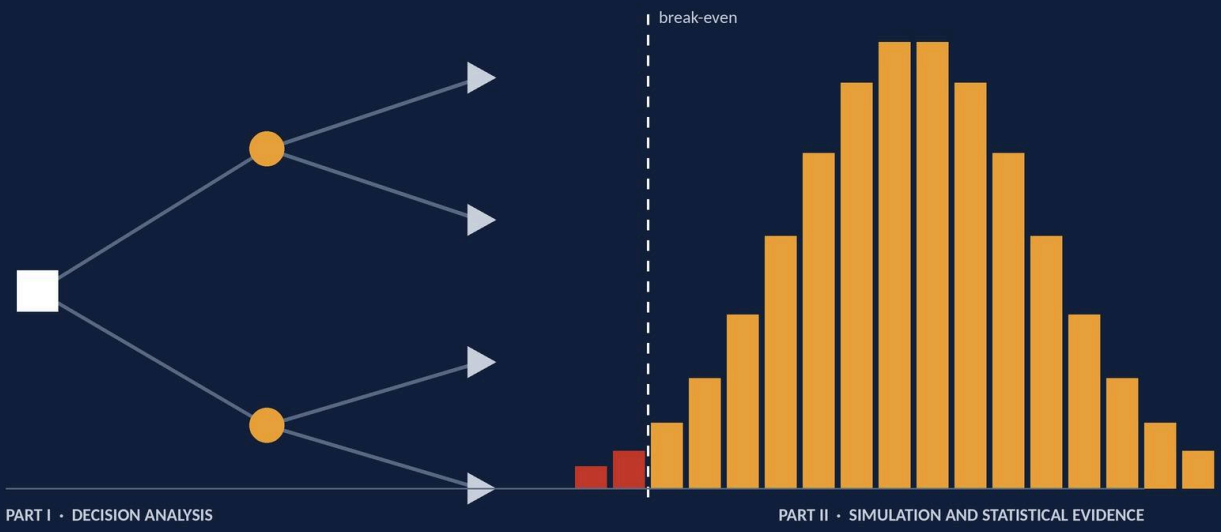
MG628 · DATA ANALYSIS FOR MANAGERIAL DECISIONS



Decisions Under Uncertainty

A Study Guide to Probability, Expected Value, and Monte Carlo Simulation

FOR STUDENTS IN

Business | Healthcare Administration | Public Administration

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Decisions Under Uncertainty: A Study Guide

MG628 · Data Analysis for Managerial Decisions

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Preface

This guide teaches you to make and defend decisions when the outcome is uncertain. It follows the two-part lecture *Decisions Under Uncertainty* slide for slide, and turns its narration into a text you can read, annotate, and return to while you complete the assignment.

Part I (Chapters 1–4) builds the decision-analysis toolkit: probability, expected monetary value, the value of information, and expected utility. Part II (Chapters 5–8) moves from a single expected value to the full range of outcomes, using Monte Carlo simulation and the statistical evidence managers rely on.

How to use this guide

- **Read a chapter before or after its video.** The figures are taken from the lecture slides, so you can move between the two easily.
- **Work every "Check your understanding" problem before reading the answers** at the end of the chapter.
- **Use Chapter 7 to check your notebook.** It lists every result your assignment should produce. The interpretation and the recommendation remain yours to write.
- **Look for the Manager's Lens** at the end of each concept. It asks three questions: what decision the tool supports, what it tells you to do, and what a careless reading gets wrong.

Three running cases

Three scenarios, one from each program track, run through Part I. Each is a bet with a good outcome and a bad one.

Track	Decision	Good outcome	Bad outcome
Business	Launch a new product?	60%: +$100,000 profit	40%: −$30,000 loss

Track	Decision	Good outcome	Bad outcome
Healthcare	Open an Express Flu Clinic?	70%: +$150,000 revenue	30%: −$50,000 loss
Public administration	Install smart streetlight sensors?	50%: +$200,000 savings	50%: −$100,000 repair

The fields differ, but the question is the same: is the likely gain worth the possible loss, and can the organization survive the loss if it happens?

Learning objectives

By the end of this guide you should be able to:

1. Apply probability to decisions in business, public administration, and healthcare.
2. Evaluate expected monetary value (EMV) as a decision criterion, including how a sure amount (the certainty equivalent) compares with a gamble.
3. Calculate the expected value of perfect information (EVPI) and of sample information (EVSI) to decide whether data is worth its cost.
4. Explain why expected utility can beat EMV for risk-averse organizations, and use utilities in a decision tree.
5. Use Monte Carlo simulation to see the full range of outcomes and their likelihoods, and support a recommendation with statistical evidence.

Part I · Decision Analysis

Chapter 1. Probability: The Language of Risk

Managers rarely lack opinions about the future; what they lack is a way to state those opinions precisely enough to plan, budget, and compare. Probability provides that language.

1.1 From vague words to numbers

A probability is a number from 0 to 1. Zero means impossible, one means certain, and 0.5 is a coin flip. Saying "70%" instead of "probably" lets a team compare risks, budget for them, and later check whether the estimate was right.

Event	Probability	Track
A "100-year" flood in a given year	1%	Public administration
A surge in the emergency department on a Friday night	15%	Healthcare
A competitor launches a similar product	60%	Business
A high-volume flu season	70%	Healthcare

Managers get probabilities from three main sources:

- **Historical frequency.** Admissions records show a surge on 15% of Friday nights.
- **Models built on data.** River gauge records put the yearly flood chance at 1%.
- **Expert judgment.** Market research rates the launch's chance of success at 60%.

Good practice is to record which source each number came from, so it can be challenged and updated.

1.2 The rules every probability must follow

1. **Between 0 and 1.** No event is less likely than impossible or more likely than certain.
2. **All outcomes sum to 1.** List every possible outcome exactly once: a high-volume season (0.7) plus a mild season (0.3) covers every case.
3. **Complement.** P(not A) = 1 − P(A). The chance of no surge on a Friday is 1 − 0.15 = 0.85.
4. **Addition rule ("or").** P(A or B) = P(A) + P(B) − P(A and B). Subtract the overlap so it is not counted twice.

Worked example (Healthcare). On a given Friday, P(surge) = 0.15, P(equipment failure) = 0.10, and P(both) = 0.03. The chance the night is disrupted by either is 0.15 + 0.10 − 0.03 = **0.22**. Simply adding 0.15 and 0.10 would double-count the 3% of nights with both.

1.3 Mutually exclusive is not the same as independent

	Meaning	Rule	Example
Mutually exclusive	Cannot happen together	P(A and B) = 0	A flu season is high-volume or mild, never both

	Meaning	Rule	Example
Independent	One tells you nothing about the other	P(A and B) = P(A) × P(B)	If surges were independent, two surge Fridays in a row: 0.15 × 0.15 = 0.0225

Independence is an assumption, and it often fails. Surges cluster in flu season. Suppose half the months run at 25% per Friday and half at 5%, still 15% on average. The chance of at least one surge in a four-Friday month is then $1 - 0.75^4 =$ **68%** in flu months and $1 - 0.95^4 =$ **19%** in quiet months.

The 48% average you will meet in Section 1.4 hides a roster that must be ready in flu months and can relax in quiet ones. Always ask whether independence fits your situation.

1.4 Small probabilities add up over time

The **complement rule** says the probability that an event does not happen is one minus the probability that it does. From it comes a rule managers use constantly: the chance that an event happens at least once in *n* independent periods.

P(not A) = 1 − P(A) and **P(at least once in n periods) = $1 - (1 - p)^n$**

Example 1.1 (Healthcare). A surge occurs on 15% of Friday nights. Over a month with four Fridays, the chance of at least one surge is $1 - 0.85^4 = 0.478$, about **48%**.

Example 1.2 (Public administration). A 100-year flood has a 1% chance in any year. Over a 30-year levee bond, the chance of at least one is $1 - 0.99^{30} = 0.260$, about **26%**.

A risk that sounds rare per event can be close to a coin flip over the period a manager is responsible for. The calculation assumes each period is independent of the others.

1.5 Conditional probability

Count first. Imagine 1,000 launches under the same conditions. A competitor launches in 600 of them; with a 45% success rate, 270 succeed and 330 fail. Of the 400 launches with no competitor, 82%, or 328, succeed and 72 fail.

	Succeed	Fail	Total
Competitor launches	270	330	600
No competitor	328	72	400

	Succeed	Fail	Total
Total	**598**	**402**	**1,000**

Every conditional probability is now a simple fraction:

- **P(succeed | competitor)** = 270 ÷ 600 = 0.45 (read across the competitor row).
- **P(fail | competitor)** = 330 ÷ 600 = 0.55 (same row, other column).
- **P(competitor | fail)** = 330 ÷ 402 = 0.82 (read down the fail column).

P(A | B) is not P(B | A). Fifty-five percent of competitor launches fail, but 82% of failures had a competitor. Confusing the two is one of the most common errors in business cases.

Probabilities depend on what else happens. The vertical bar in P(B | A) is read "given": the chance of B once we know A happened. The **multiplication rule** combines them.

P(A and B) = P(A) × P(B | A)

Example 1.3 (Business). There is a 60% chance a competitor launches a similar product. If it does, our product succeeds with probability 0.45; if not, with probability 0.82. Adding the two success paths gives the overall chance of success.

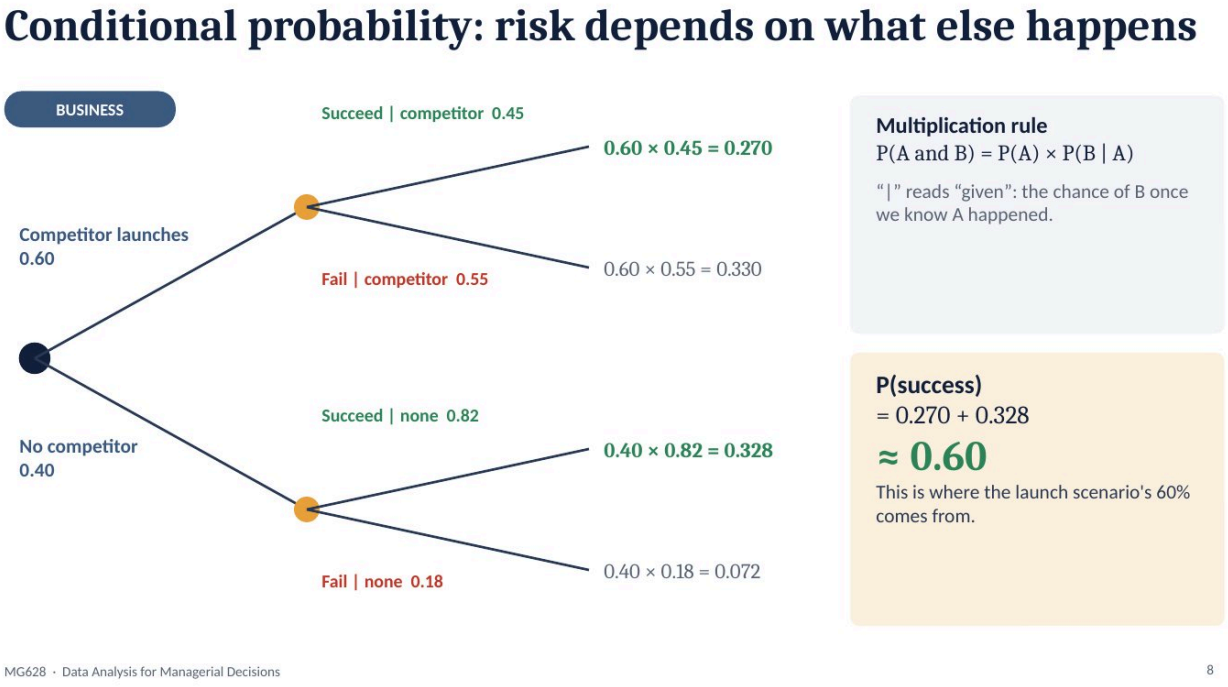


Figure 1.1. Each path's probability is the product of its branches. The two success paths, 0.270 and 0.328, sum to about 0.60.

That is where the launch scenario's 60% comes from. When you hear a single probability in a business case, ask what it was conditioned on.

1.6 Manager's Lens: probability

> **The decision it supports.** How much capacity, inventory, insurance, or protection to buy against an uncertain event.
>
> **What it tells you to do.** Size the response to the probability over your planning horizon (the month, the budget year, the bond term), not per event.
>
> **What a careless reading gets wrong.** Reading "only 15%" as "it won't happen," or ignoring how a competitor's move changes your own odds.

- **Business:** a 60% chance of a competing product is a reason to stage R&D spending, not to cancel it.
- **Healthcare:** a 48% monthly surge chance justifies a standing on-call roster rather than ad hoc overtime.
- **Public administration:** a 26% flood chance over the bond's life makes levee height a budget decision, not an engineering detail.

Chapter 1 summary

- A probability converts "it might happen" into a number from 0 to 1 that a team can plan around.
- Over many periods, the chance of at least one occurrence is $1 - (1 - p)^n$, and it grows quickly.
- Conditional probabilities combine through the multiplication rule; a headline probability is only as good as what it was conditioned on.

Chapter 2. Expected Monetary Value

Once outcomes have probabilities and payoffs, a risky option can be summarized by one number, its expected monetary value. This chapter shows how to compute it and, just as important, what it leaves out.

2.1 EMV: the probability-weighted average payoff

Start with counts. Picture the same launch decision made 100 times. About 60 succeed and 40 fail.

Count × payoff	Dollars
60 × +$100,000	+$6,000,000
40 × −$30,000	−$1,200,000
Total for 100 launches	$4,800,000
Average per launch	**$48,000**

Divide every count by 100 and the counts become probabilities: 0.6 × $100,000 + 0.4 × (−$30,000) = $48,000. The expected value formula is simply this long-run average, written with probabilities instead of counts.

Multiply each outcome's payoff by its probability, then add the results.

EMV = $\Sigma\, p_i \times x_i$, summed over every outcome i, where p_i is its probability and x_i its payoff.

Keep three properties in mind:

1. **The probabilities must sum to 1**, so every possible outcome is listed exactly once.
2. **Losses enter as negative numbers.** A cost or repair bill reduces the EMV.
3. **EMV is a long-run average.** It describes many repetitions of a decision, not the result of any single one.

2.2 Worked example: should the company launch?

Outcome	Probability	Payoff	Probability × payoff
Success	0.60	+$100,000	+$60,000
Failure	0.40	−$30,000	−$12,000
EMV			**$48,000**

Launching ($48,000) beats not launching ($0), so a decision-maker who uses EMV alone launches. Notice that $48,000 is not one of the possible results: the company will either earn $100,000 or lose $30,000.

2.3 Three scenarios, three downsides

The same calculation gives an EMV of $48,000 for the launch, 0.7($150,000) + 0.3(−$50,000) = $90,000 for the flu clinic, and 0.5($200,000) + 0.5(−$100,000) = $50,000 for the smart-city sensors. All three are better on average than doing nothing.

Scenario	Best case	EMV	Worst case
Product launch	+$100,000	$48,000	−$30,000
Flu clinic	+$150,000	$90,000	−$50,000
Smart-city sensors	+$200,000	$50,000	−$100,000

The smart-city project has a higher EMV than the launch, yet its worst case is more than three times larger. EMV ranks options by their average, not by how badly they can go wrong; a manager has to look at both.

2.4 The EMV is a long-run average, not a forecast

Figure 2.1 simulates many product launches and tracks the running average profit. Early on the average jumps around; as launches accumulate it settles toward $48,000.

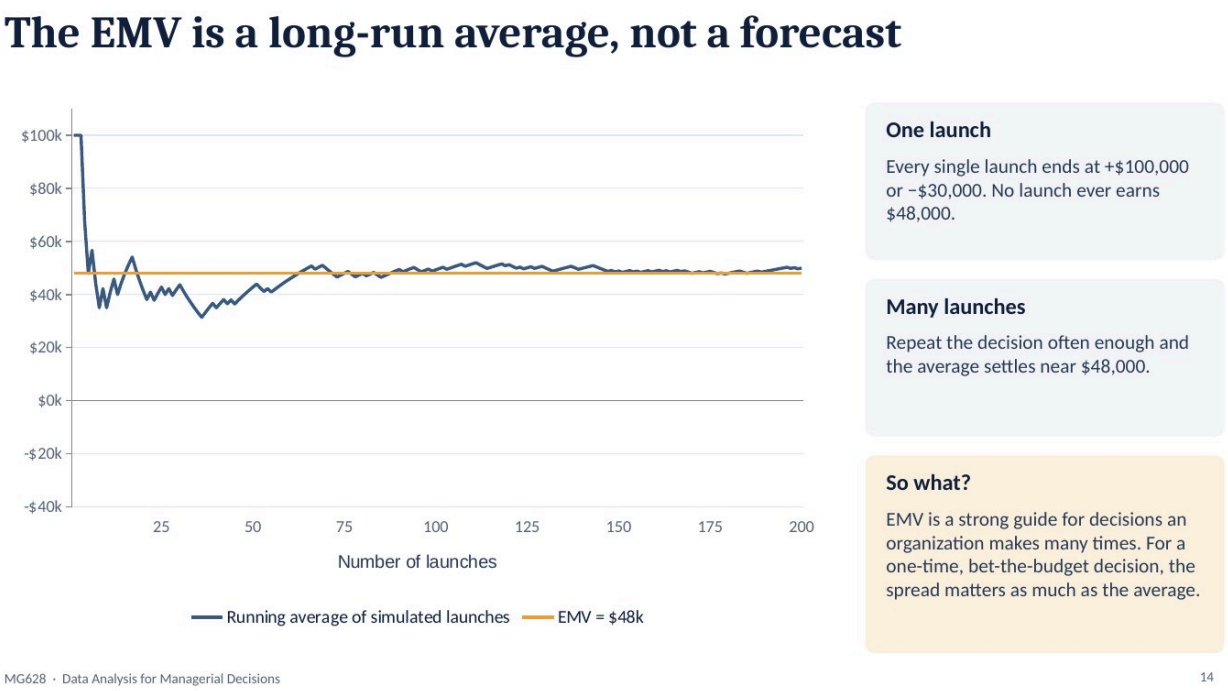


Figure 2.1. Each single launch ends at +$100,000 or −$30,000. Only the average over many launches approaches $48,000.

EMV is a strong guide for decisions an organization makes many times. For a one-time, bet-the-budget decision, the spread of outcomes matters as much as the average. This simulation is also a first look at the Monte Carlo approach of Part II.

2.5 Measuring risk: the standard deviation of a decision

EMV gives the center of a decision's outcomes. The **standard deviation (SD)** measures how far a single outcome typically lands from that center.

Variance = $\Sigma\, p_i \times (x_i - \text{EMV})^2$, and **SD = $\sqrt{\text{variance}}$**.

Example 2.1 (Business).

Outcome	p	x − EMV	p × (x − EMV)²
Success	0.6	+$52,000	$1.62 billion
Failure	0.4	−$78,000	$2.43 billion
Variance			**$4.06 billion**

SD = $\sqrt{4.06 \text{ billion}} \approx$ **$63,700**.

Scenario	EMV	SD
Product launch	$48,000	$63,700
Flu clinic	$90,000	$91,700
Smart-city sensors	$50,000	$150,000

The sensors and the launch have similar EMVs, but the sensors' SD is more than twice as large. Same average, far more risk: the risk is now a number, not an adjective. Part II relies on this same measure.

2.6 How wrong can the probability be? Break-even analysis

Probabilities are estimates, so ask how wrong an estimate can be before the decision changes. Launching is worth it on average when $100p - 30(1 - p) > 0$, which means **$p > 0.23$**.

Scenario	Estimated probability	Break-even probability
Product launch	60%	23%
Flu clinic	70%	25%
Smart-city sensors	50%	33%

The gap between the estimate and the break-even is a **margin of safety**. The launch estimate could fall from 60% to 23% before the EMV decision changes. The sensors have the least room: 50% against 33%.

2.7 Sure thing or gamble? The certainty equivalent

Suppose you can take **Option A**, a guaranteed $45,000, or **Option B**, a 50/50 gamble between $100,000 and $0. Option B's EMV is $50,000, so an EMV decision-maker chooses B. Many real managers choose A anyway.

The guaranteed amount that feels exactly as good as the gamble is the **certainty equivalent (CE)**. The gap between the EMV and the CE is the **risk premium**, the expected value you give up for certainty.

Risk premium = EMV − CE = $50,000 − $45,000 = $5,000

2.8 Your certainty equivalent reveals your risk attitude

Attitude	Rule	Behavior	Example
Risk-averse	CE < EMV	Gives up some average payoff to avoid the swing	A hospital already running a deficit; a city spending scarce taxpayer funds
Risk-neutral	CE = EMV	Cares only about the average; this is what "decide by EMV" assumes	A large firm making many similar, independent bets each year
Risk-seeking	CE > EMV	Pays for a chance at the big upside	A startup that must hit a large target to survive another funding round

Choosing the highest EMV is choosing to act risk-neutral. That is a management decision, so make it deliberately rather than by default.

2.9 Manager's Lens: EMV and the certainty equivalent

> **The decision it supports.** Ranking options when every option has uncertain outcomes.
>
> **What it tells you to do.** A positive EMV means the bet pays on average. The certainty equivalent tells you whether this organization, in its current situation, should take it.
>
> **What a careless reading gets wrong.** Treating EMV as the expected result of a one-time decision, or never asking whether the worst case is survivable.

- **Business:** a firm that launches many products can absorb a $30,000 miss; EMV is a sound guide here.
- **Healthcare:** in a budget deficit, a $50,000 loss may force service cuts, so the hospital's certainty equivalent for the clinic can be far below $90,000.
- **Public administration:** a $100,000 taxpayer loss also costs public trust, a cost the EMV does not include.

Check your understanding 2.1

A city council can install the smart sensors (50%: save $200,000; 50%: pay $100,000 for repairs) or skip the project. The council says it would be equally happy with the gamble or a guaranteed $40,000 from a utility rebate program.

1. What is the EMV of installing the sensors?
2. What are the council's certainty equivalent and risk premium?
3. Is the council risk-averse, risk-neutral, or risk-seeking?

Answers. (1) 0.5 × $200,000 + 0.5 × (−$100,000) = **$50,000**. (2) The CE is **$40,000**, so the risk premium is $50,000 − $40,000 = **$10,000**. (3) The CE is below the EMV, so the council is **risk-averse**. It is paying $10,000 in expected value for certainty; with public money and public trust at stake, that can be a defensible price.

Chapter 2 summary

- EMV is the probability-weighted average payoff, and it is a long-run concept.
- Two options with good EMVs can have very different worst cases.
- The certainty equivalent and risk premium measure how much an organization will pay to avoid risk; deciding by EMV means acting risk-neutral.

Chapter 3. The Value of Information

Before committing to a risky decision, a manager can often pay to reduce the uncertainty. This chapter shows how to calculate what that information is worth, so you never pay more for it than the decision can recover.

3.1 Information is something you buy

Track	Information	Question it answers
Business	Market study	Will customers actually buy the product?
Healthcare	Diagnostic test	Does this patient need the expensive treatment?
Public administration	Pilot program	Will the sensors work with the city's old wiring?

The decision rule is simple: **buy information only if its expected value is greater than its cost.** Information has value only when it could change what you decide. Two measures put a number on that value:

- **EVPI**, the expected value of perfect information: the value of a perfect forecast, and the most any information could be worth.
- **EVSI**, the expected value of sample information: the value of real, imperfect information such as a survey, test, or pilot.

3.2 EVPI in three steps

EVPI = EV with perfect information − EMV of the best decision without it

Example 3.1 (Business).

1. **Best decision without information.** Launch: 0.6 × \$100,000 + 0.4 × (−\$30,000) = **\$48,000**.
2. **Expected value with perfect information.** If the forecast says success, launch (+\$100,000); if it says failure, don't launch (\$0). So 0.6 × \$100,000 + 0.4 × \$0 = **\$60,000**.
3. **EVPI** = \$60,000 − \$48,000 = **\$12,000**.

The \$12,000 has a plain meaning: perfect information lets you avoid the \$30,000 loss in the 40% of cases where the launch would fail (0.4 × \$30,000).

A second route: expected regret. Regret is what you would have earned with the best choice for an outcome, minus what you actually earned.

Decision	Regret if success (0.6)	Regret if failure (0.4)	Expected regret
Launch	$0	$30,000	0.4 × $30,000 = **$12,000**
Don't launch	$100,000	$0	0.6 × $100,000 = $60,000

The smallest expected regret, $12,000, equals the EVPI. Two routes give one answer, for a good reason: perfect information removes regret entirely, so the most it can be worth is the regret you expect to carry under your best decision without it.

3.3 EVPI is largest where uncertainty hurts most

Scenario	EV with perfect information	EMV	EVPI
Product launch	$60,000	$48,000	$12,000
Flu clinic	$105,000	$90,000	$15,000
Smart-city sensors	$100,000	$50,000	$50,000

The sensors stand out. A 50/50 chance and a $100,000 downside mean a forecast would change the decision half the time, so a pilot could be worth up to $50,000, more than four times the value of the launch study. Information is most valuable when uncertainty is high and being wrong is expensive.

3.4 Real information is imperfect: Bayes' rule

Example 3.2 (Public administration). The city can test the sensors on a few streetlights first. The pilot gives the right answer 90% of the time.

Count first. Imagine 1,000 situations like this one. In 500 the technology works and in 500 it fails.

	Pilot passes	Pilot fails	Total
Technology works	450	50	500
Technology fails	50	450	500
Total	**500**	**500**	**1,000**

Suppose the pilot passed. Look only at the 500 passes: 450 of them work, so the chance the technology works is 450 ÷ 500 = 0.90. After a failed pilot it is 50 ÷ 500 = 0.10. Figure 3.1 shows the same counts as probabilities.

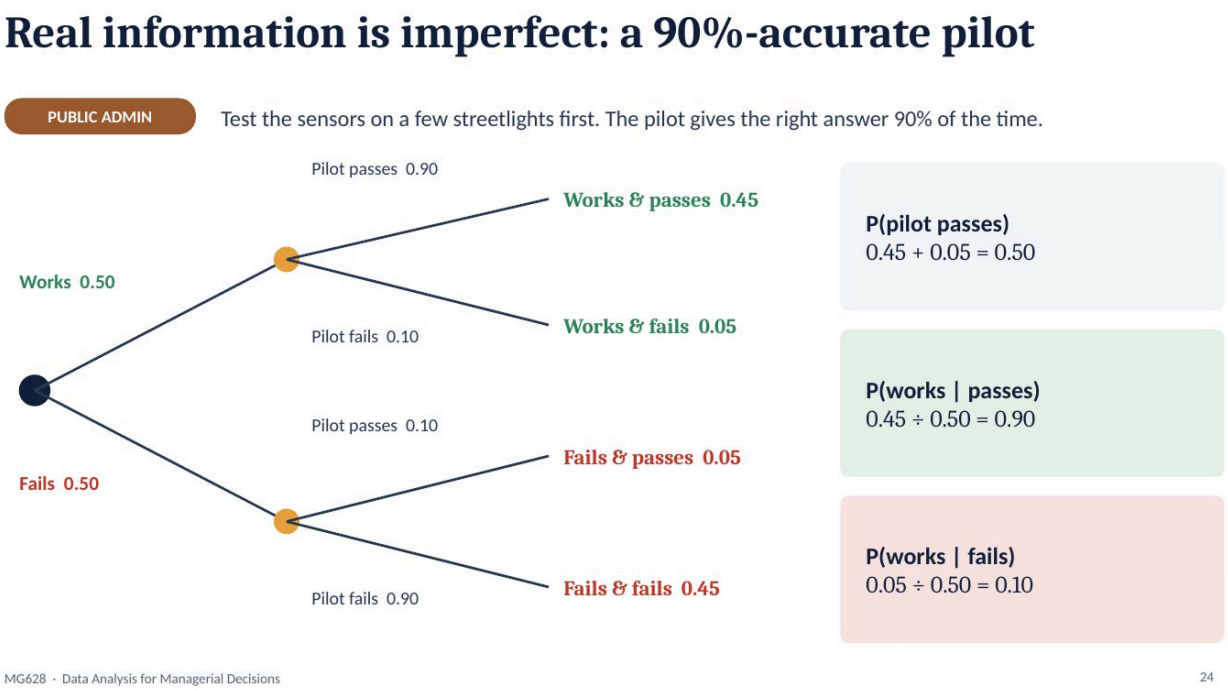


Figure 3.1. Multiplying along each path gives four joint probabilities. The pilot passes with probability 0.45 + 0.05 = 0.50.

The pilot result revises the chance that the technology works. This is **Bayes' rule**: new evidence updates a probability.

P(works | passes) = 0.45 ÷ 0.50 = 0.90 and **P(works | fails) = 0.05 ÷ 0.50 = 0.10**

3.5 EVSI: what the pilot is actually worth

- **If the pilot passes (50%),** install: 0.9 × $200,000 − 0.1 × $100,000 = **$170,000**.
- **If the pilot fails (50%),** installing would average 0.1 × $200,000 − 0.9 × $100,000 = −$70,000, so don't install: **$0**.

Step	Calculation	Value
Expected value with the pilot	0.5 × $170,000 + 0.5 × $0	$85,000
Minus EMV without the pilot	install now	$50,000
EVSI		**$35,000**

Run the pilot only if it costs less than **$35,000**. Its efficiency is EVSI ÷ EVPI = $35,000 ÷ $50,000 = 70%: imperfect information is worth less than perfect information, but it can still be well worth buying.

The full decision as one tree. Figure 3.2 puts every choice in order: run the pilot or not, see the result, then decide whether to install.

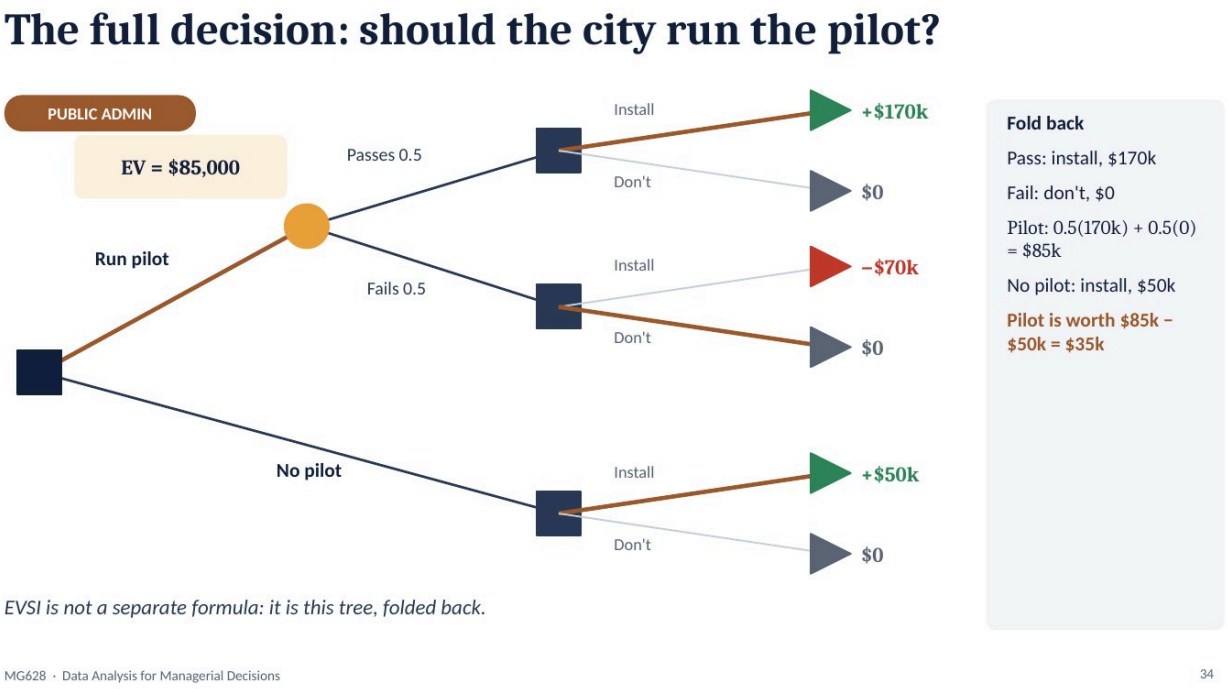


Figure 3.2. After a pass the city installs ($170,000); after a fail it does not ($0). The pilot branch is worth 0.5($170,000) + 0.5($0) = $85,000, against $50,000 for installing without it.

The pilot is worth $85,000 − $50,000 = $35,000 before its cost. EVSI is not a separate formula: it is this tree, folded back.

3.6 Manager's Lens: the value of information

> **The decision it supports.** Whether to pay for a study, test, or pilot before committing resources.
>
> **What it tells you to do.** EVPI caps what any information can be worth. EVSI tells you what a particular study is worth. Buy only if EVSI exceeds the cost.
>
> **What a careless reading gets wrong.** "More data is always better," which leads to paying more for research than the decision can ever recover.

- **Business:** never pay more than $12,000 for launch research, however impressive the vendor's pitch.
- **Healthcare:** a test is worth ordering only if a result could change the treatment or staffing decision.
- **Public administration:** a pilot worth up to $35,000 also lowers the political risk of a large, visible failure.

Chapter 3 summary

- Information has value only if it could change the decision.
- EVPI = EV with perfect information − EMV; it is the ceiling on what any study is worth.
- Bayes' rule revises probabilities after imperfect evidence; EVSI values that evidence and sets the most you should pay for it.

Chapter 4. Expected Utility and Decision Trees

The same dollar loss can hurt some organizations far more than others. Expected utility builds that difference into the decision, and the decision tree is where you apply it.

4.1 Anatomy of a decision tree

A decision tree lays out a decision visually with three kinds of nodes:

- **Decision node (square):** a choice the manager controls.
- **Chance node (circle):** an uncertain event, with probabilities on its branches.
- **End point (triangle):** the payoff for that path, in dollars or in utility.

To solve a tree, **fold it back from right to left**. At each chance node, take the probability-weighted average of its branches. At each decision node, choose the branch with the best value. The value that reaches the root is the value of the best strategy.

4.2 The flu clinic, folded back with EMV

Example 4.1 (Healthcare). The hospital can open the clinic or not. Opening gives a 70% chance of +$150,000 and a 30% chance of −$50,000; not opening is worth $0.

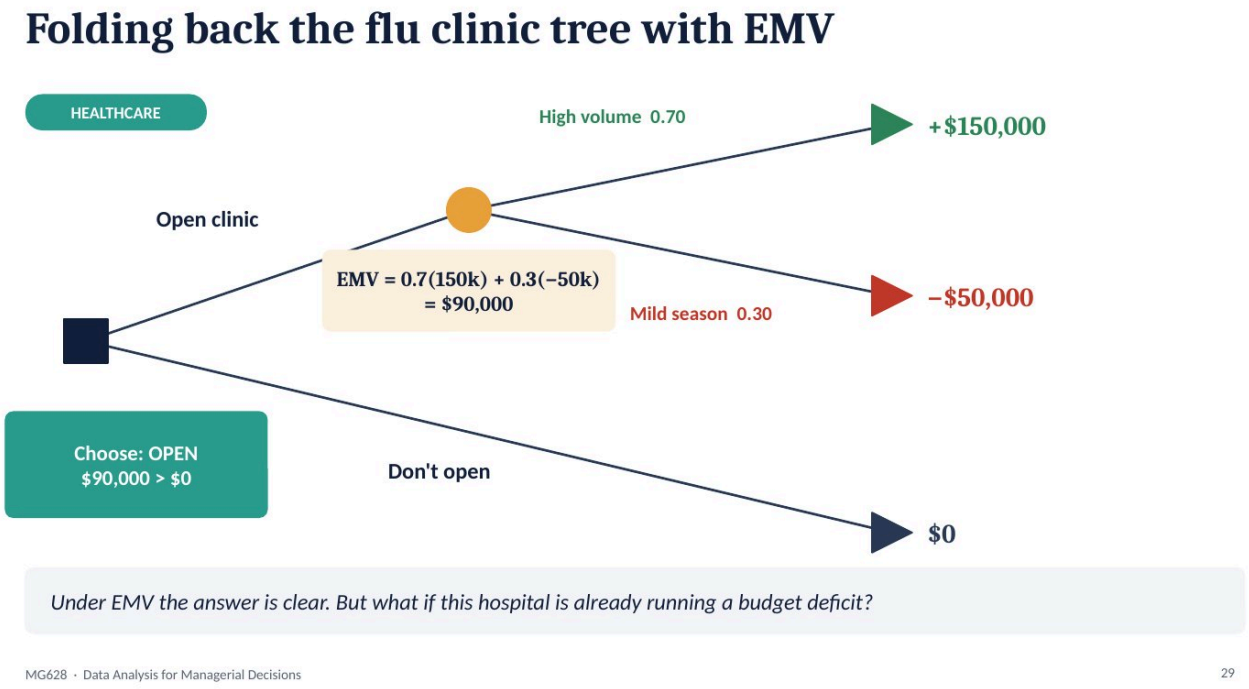


Figure 4.1. The chance node is worth 0.7($150,000) + 0.3(−$50,000) = $90,000, which beats $0. Under EMV, the hospital opens.

Under EMV the answer is clear. But what if this hospital is already running a budget deficit?

4.3 Utility: why dollars are not the whole story

Utility is a score for how good or bad an outcome is for a particular organization, not just its dollar amount. The utility values come from management: they encode how much the organization can bear.

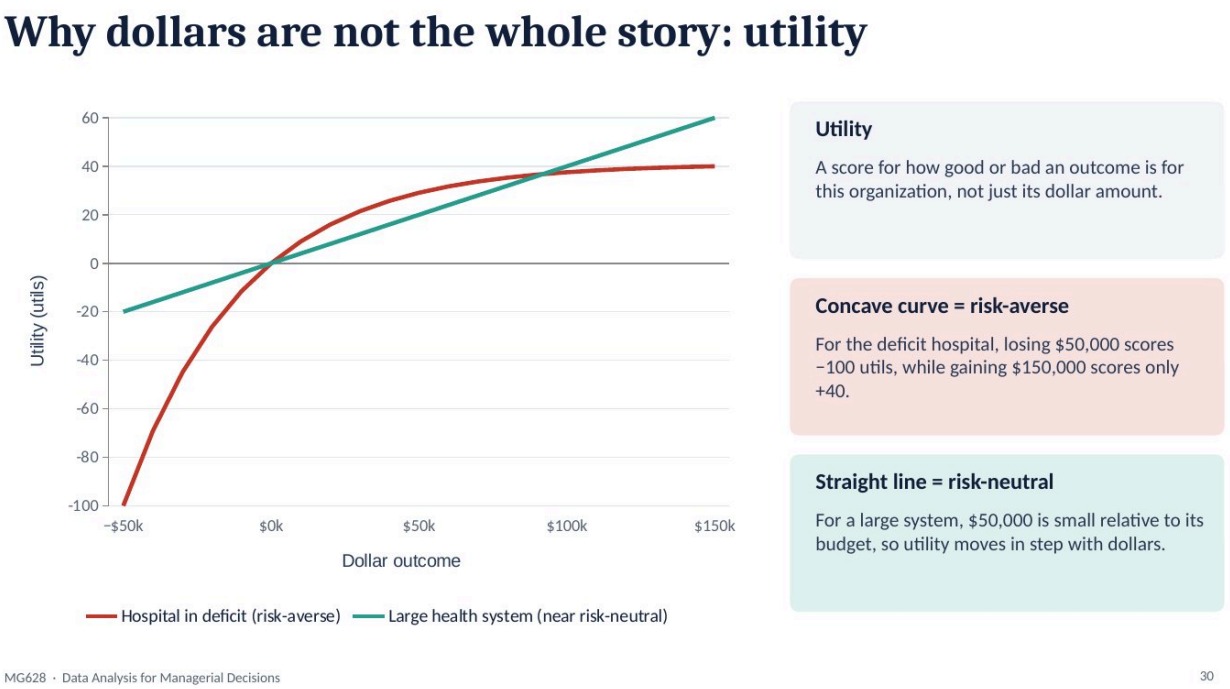


Figure 4.2. A concave curve means risk-averse; a straight line means risk-neutral.

For the hospital in a deficit (the concave curve), losing $50,000 scores −100 utils, while gaining $150,000 scores only +40. For a large, well-funded system (the straight line), $50,000 is small relative to its budget, so utility moves in step with dollars.

4.4 Where utilities come from: the lottery question

Utilities are not invented; they come from a question a manager can answer. Offer the decision-maker $0 for sure, or a lottery that pays +$150,000 with probability p and −$50,000 otherwise. Ask: **at what p are you indifferent?**

Set the best outcome at a utility of 1 and the worst at 0. The answer p is then the utility of $0 on that 0-to-1 scale.

Organization	Answer	What it reveals
Hospital in a budget deficit	p = 0.714	Demands a 71% chance of winning before risking a loss it cannot absorb
Large health system	p = 0.25	Accepts the bet whenever its average is positive (150p = 50(1 − p)): risk-neutral

Rescaled, these answers give exactly the utilities used in Section 4.5: −100, 0, and +40 utils for the deficit hospital, and −20, 0, and +60 for the large system. They also connect back to Chapter 2: on the deficit hospital's curve, the 70% clinic gamble is worth about **−$1,900 for sure**, despite an EMV of $90,000.

4.5 Expected utility: same gamble, different organizations

Expected utility (EU) works like EMV, with utilities in place of dollars.

$EU = \Sigma\, p_i \times U(x_i)$, where $U(x_i)$ is the utility of outcome i.

Outcome	Hospital in deficit	Large health system
+$150,000 (high volume, 0.7)	+40 utils	+60 utils
−$50,000 (mild season, 0.3)	−100 utils	−20 utils
$0 (don't open)	0 utils	0 utils
EU(open)	0.7(40) + 0.3(−100) = **−2**	0.7(60) + 0.3(−20) = **+36**
Decision	**Don't open** (0 > −2)	**Open** (+36 > 0)

Same gamble, same probabilities, same dollars, and the right decision differs because the organizations differ.

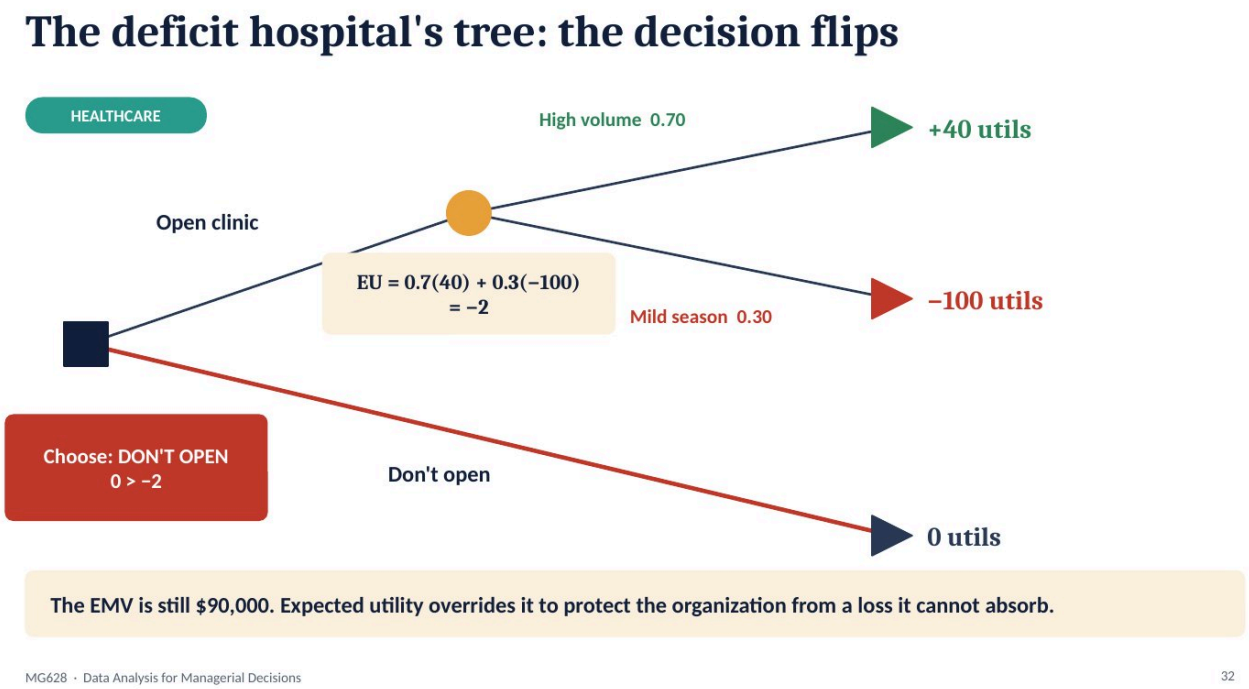


Figure 4.3. With utilities at the end points, the chance node is worth −2 and the decision flips to "don't open."

The EMV of opening is still $90,000. Expected utility overrides it to protect the organization from a loss it cannot absorb. In practice: replace the dollar values at the ends of the branches with utilities, compute EU at the chance nodes, and choose the branch with the highest EU, even if its EMV is lower.

4.6 Manager's Lens: risk tolerance and catastrophic downside

> **The decision it supports.** Choosing between projects with similar EMVs but very different worst cases.
>
> **What it tells you to do.** Write down the organization's risk tolerance, score outcomes in utility, and pick the highest expected utility. Rule out any option whose worst case threatens survival or mission.
>
> **What a careless reading gets wrong.** Assuming two projects with the same EMV are equally good.

- **Business:** one failure costs a quarter's margin; another ends the product line. Same EMV, very different risk.
- **Healthcare:** a clinic that strains a deficit budget can lead to staff cuts, longer wait times, and lower morale.
- **Public administration:** a visible taxpayer loss can cost public trust and reappointment long after the dollars are recovered.

Check your understanding 4.1

1. **Business.** A research firm offers a perfectly accurate forecast for the product launch at a price of $15,000. Should the company buy it? What is the most it should ever pay?
2. **Healthcare.** The deficit hospital scores +$150,000 at +40 utils and −$50,000 at −100 utils. Let p be the probability of a high-volume season. How large must p be before opening the clinic is worthwhile?

Answers. (1) **Don't buy it.** The EVPI is $12,000, so a perfect forecast is worth at most $12,000; at $15,000 the company would expect to lose $3,000. (2) Open only if $40p - 100(1 - p) > 0$, so $140p > 100$ and $\boldsymbol{p > 0.714}$**, about 71%**. At 70% the hospital is just short of the threshold. When a decision sits this close to its threshold, better information about the season can change it, which is exactly when information is worth buying.

4.7 Your decision-analysis toolkit

Tool	Question it answers	Use it when
Probability	How likely is each outcome over my planning horizon?	Sizing capacity, inventory, or protection
EMV	Which option pays best on average?	Repeated decisions with survivable downsides
Standard deviation	How far from the EMV can one outcome land?	Comparing options with similar averages
Break-even probability	How wrong can my estimate be before the decision flips?	Probabilities are estimates, not facts
Certainty equivalent	What sure amount would I accept instead of the gamble?	Comparing a guaranteed option with a risky one
EVPI / EVSI	Is more information worth its cost?	Before paying for a study, test, or pilot
Expected utility	Which option is best given our risk tolerance?	One-time decisions with severe worst cases

Chapter 4 summary

- Decision trees are solved by folding back: average at chance nodes, choose at decision nodes.
- Utility scores outcomes by how much they matter to a specific organization; a concave curve signals risk aversion.
- Expected utility can rightly reject an option with a higher EMV when its worst case is more than the organization can bear.

Part II · Simulation and Statistical Evidence

Chapter 5. Monte Carlo Simulation

Part I worked with one uncertain event and two outcomes, which gave a single EMV from a tree you could draw by hand. Real decisions depend on many uncertain inputs at once: demand, prices, and holding costs; patient volume, overtime, and reimbursements;

participation, overhead, and program benefits. That is too many branches for a tree, so we simulate instead.

Monte Carlo simulation answers questions an average cannot: "How often do we lose money?" and "What is the realistic worst case?"

5.1 Four steps of a Monte Carlo simulation

1. **Build the model.** Write the formula that turns inputs into the outcome you care about, such as profit or net margin.
2. **Describe the uncertainty.** Give each uncertain input a probability distribution instead of a single guess.
3. **Draw and repeat.** Draw random input values, compute the outcome, and repeat thousands of times.
4. **Analyze the outcomes.** Study the distribution: its average, its spread, and how often things go wrong.

The payoff for managers is a better kind of statement. Instead of "the average cost is $X," you can say "there is an 85% chance we stay under budget."

5.2 The practice model: how many units to stock

The practice dataset comes from an inventory decision.

Input	Value
Order quantity	500 units
Fixed cost of the order	$5,000
Selling price	$20 per unit
Holding cost	$2 per unsold unit
Demand (uncertain)	Normal, mean 450, standard deviation 100

Profit = 20 × min(demand, 500) − 5,000 − 2 × unsold units

Demand	What happens	Calculation	Profit
600	Sells all 500; nothing unsold	20(500) − 5,000 − 0	+$5,000

Demand	What happens	Calculation	Profit
436	Sells 436; 64 unsold	20(436) − 5,000 − 2(64)	+$3,592
250	Sells 250; 250 unsold	20(250) − 5,000 − 2(250)	−$500

Demand is the only uncertain input, so every simulated period differs only in its demand draw.

5.3 The normal distribution behind uncertain demand

The practice model describes demand with a **normal distribution**: the bell curve, defined by its mean μ (the center, 450 units) and standard deviation σ (the typical distance from the center, 100 units).

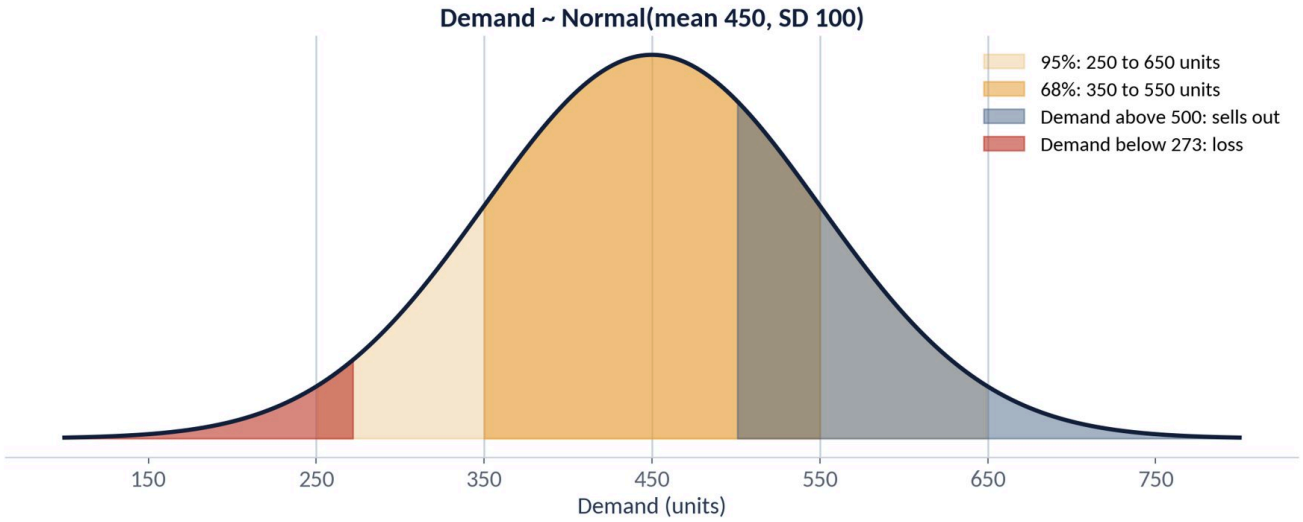


Figure 5.1. About 68% of demand falls between 350 and 550 units and 95% between 250 and 650. The tails are what a manager worries about: selling out (above 500) and losing money (below about 273).

The 68–95–99.7 rule. About 68% of values fall within 1 SD of the mean, 95% within 2 SD, and 99.7% within 3 SD.

z-scores turn a threshold into a probability. A z-score measures how many standard deviations a value sits from the mean: **$z = (x - \mu) \div \sigma$**.

Threshold	z-score	Probability from the normal curve	Simulation found
Sell-out: demand above 500	(500 − 450) ÷ 100 = 0.50	P(z > 0.50) = 30.9%	30.6%
Loss: demand below 272.7	(272.7 − 450) ÷ 100 = −1.77	P(z < −1.77) = 3.8%	3.95%

A loss requires 22 × demand − 6,000 < 0, so demand below about 272.7 units. Theory and simulation agree to within simulation noise. With one uncertain input the normal curve alone gives the answer; with many interacting inputs only simulation can, which is why we simulate.

5.4 The same model in Python

```
import numpy as np

rng = np.random.default_rng(42)
demand = rng.normal(450, 100, 10_000)   # 10,000 possible periods

sales  = np.minimum(demand, 500)         # cannot sell more than we stocked
unsold = np.maximum(500 - demand, 0)
profit = 20*sales - 5000 - 2*unsold

print(profit.mean())         # average profit
print((profit < 0).mean())   # probability of a loss
```

The last line works because Python treats True as 1 and False as 0, so the mean of a True/False column is the share of True values. The practice file `montecarlo.csv` holds 10,000 periods generated from exactly this model, so you can analyze the output without writing the simulation yourself.

5.5 What 10,000 simulated periods look like

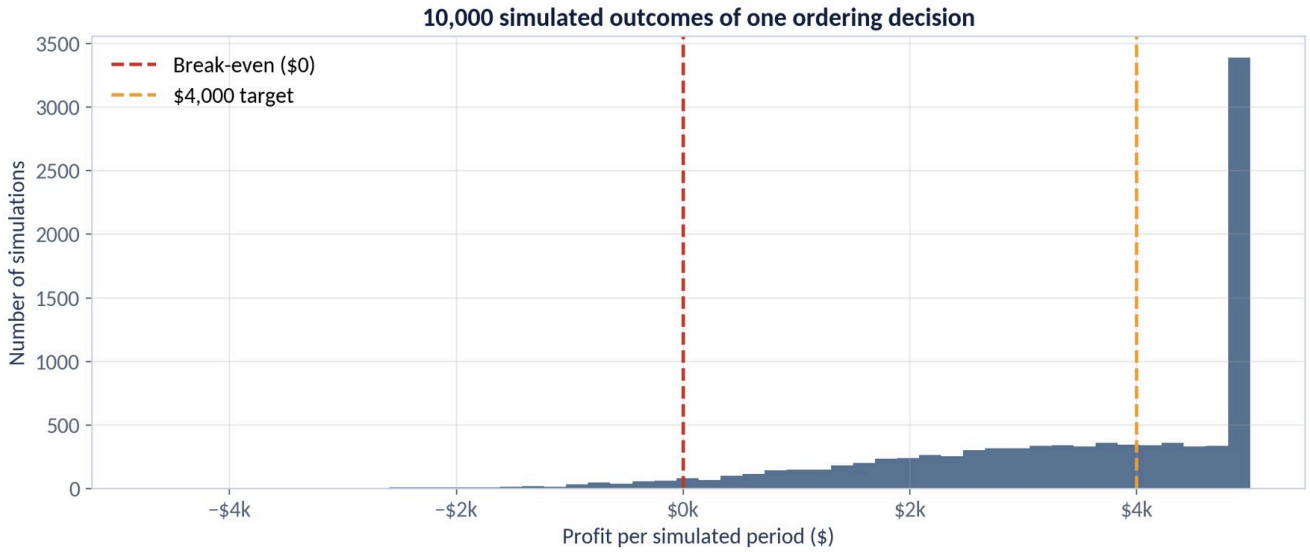


Figure 5.2. The tall bar at $5,000 is every period where demand reached 500: profit cannot go higher. The thin left tail holds the losses.

Measure	Value
Average profit	$3,460
Median profit	$3,894
Standard deviation	$1,642
Probability of a loss	3.95%
Probability of profit above $4,000	48.0%
Periods that sell out	30.6%

A single expected value would hide all of this.

5.6 Why simulation works: the law of large numbers

The table tracks the estimated probability of a loss as more and more periods are simulated.

Draws	Estimated P(loss)
10	10.0%

Draws	Estimated P(loss)
100	3.0%
1,000	3.80%
10,000	3.86%
100,000	3.88%

The law of large numbers: as the number of draws grows, a simulated average or percentage settles on the true value. The error shrinks with the square root of n, so four times as many draws halves the noise. That is why 10,000 runs is a common choice: precise enough for a management decision, fast enough to run in seconds.

5.7 Manager's Lens: Monte Carlo simulation

> **The decision it supports.** Any decision whose result depends on several uncertain inputs: stocking, staffing, budgeting, program design.
>
> **What it tells you to do.** Report the range and the odds, not just the average: "a 4% chance of loss, and about half the time we beat $4,000."
>
> **What a careless reading gets wrong.** Quoting only the average, or trusting the output without checking the model's assumptions. Bad inputs give confident-looking bad answers.

- **Business:** the sell-out rate tells you whether a larger order is worth its holding risk.
- **Healthcare:** simulating patient volume lets you staff for busy days without paying for idle shifts.
- **Public administration:** simulating participation shows a council the chance a program falls short of break-even.

Chapter 5 summary

- A Monte Carlo simulation replaces single guesses with distributions and produces thousands of possible outcomes.
- The output is a distribution, so you can report odds and ranges, not just an average.
- The results are only as good as the model and its assumptions.

Chapter 6. Reading Simulation Results

This chapter works through a complete analysis of the practice data, step by step, so you can see what a strong interpretation looks like before you write your own. Each step ends with a model interpretation: the number is stated, then explained in terms of what it means for the decision.

6.1 Six questions to ask of any simulation

#	Measure	Question it answers
1	Mean	What is the typical result?
2	Standard deviation	How predictable is it?
3	Probability of a bad outcome	How often do we fall below break-even?
4	t-test and p-value	Is the difference between options real, or chance?
5	95% confidence interval	What is the likely range for the true average?
6	Correlation and charts	What drives the risk, and what does the spread look like?

Your assignment asks these same six questions, in this order.

6.2 Standard deviation by hand

Take five simulated profits: $5,000, $3,600, $4,200, $2,400, and $1,800.

Period	Profit	Deviation from mean	Deviation squared
1	$5,000	+$1,600	2,560,000
2	$3,600	+$200	40,000
3	$4,200	+$800	640,000
4	$2,400	−$1,000	1,000,000
5	$1,800	−$1,600	2,560,000
Sum	**$17,000**	**$0**	**6,800,000**

1. **Mean:** $17,000 ÷ 5 = $3,400.
2. **Square each deviation**, which makes every distance positive.
3. **Divide by n − 1:** 6,800,000 ÷ 4 = 1,700,000. This is the variance.
4. **Take the square root:** √1,700,000 ≈ **$1,304**.

Why n − 1? A sample sits a little closer to its own mean than to the true mean, so dividing by n − 1 corrects the slight underestimate. pandas' `.std()` does this for you.

6.3 Center, spread, and risk by group

The practice notebook splits the 10,000 periods at median demand into two groups of 5,000.

Group	Mean profit	Standard deviation	Losses
High demand	$4,782.86	$337.28	0 (0.0%)
Low demand	$2,136.37	$1,332.90	395 (7.9%)
All periods	$3,459.62		

Model interpretation. High-demand periods average $4,783 with a standard deviation of only $337, and none of the 5,000 lose money. Low-demand periods average $2,136, are about four times as variable, and lose money 7.9% of the time. Every loss comes from low-demand periods, so the risk in this decision is a demand risk.

6.4 Averages vary too: the standard error

Draw 150 periods and average them; draw another 150 and the average shifts a little. Repeat 1,000 times and you get the **sampling distribution** of the mean.

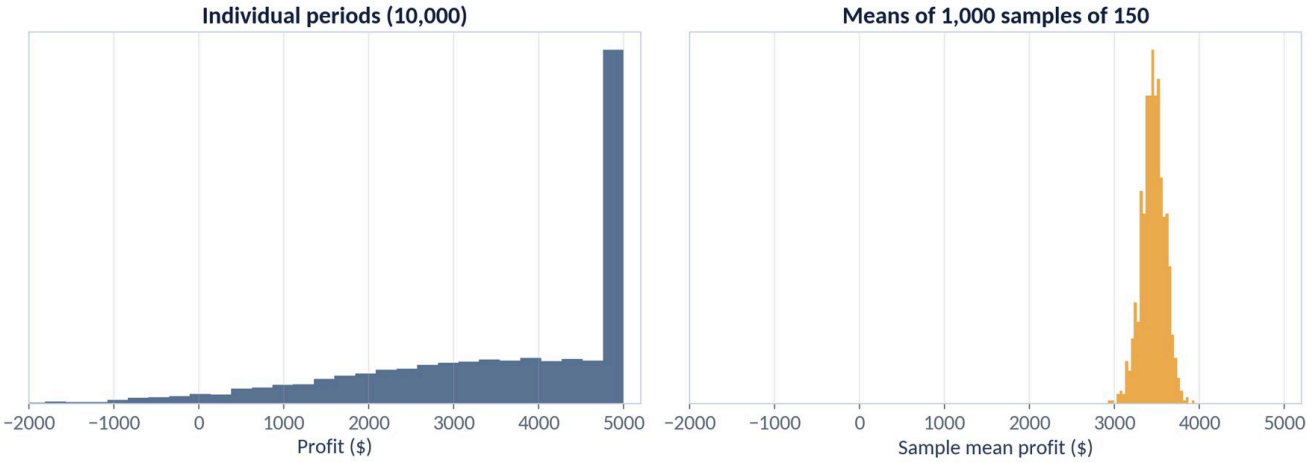


Figure 6.1. Left: 10,000 individual periods. Right: the means of 1,000 samples of 150. Averages vary far less than single periods.

Measure	Value
SD of single periods	$1,642
SD of 1,000 sample means	$138
σ ÷ √150	$134

The spread of sample averages is the **standard error: SE = s ÷ √n**. With n = 150 it is about $134, far smaller than the $1,642 spread of single periods. Every t-test and confidence interval in your assignment is built on this one idea.

6.5 Is the difference real? The two-sample t-test

- **Null hypothesis (H_0):** the two groups have the same true average; any gap we see is chance.
- **t-statistic:** how many standard errors apart the two sample means are. The larger it is in size, the stronger the evidence.
- **p-value:** if H_0 were true, the chance of seeing a gap at least this large.

Decision rule. If $p < 0.05$, reject H_0: the difference is statistically significant. If $p \geq 0.05$, fail to reject H_0: the evidence is not strong enough to say one option is better on average, even if its mean is higher.

Building the t-statistic. The t-statistic asks how many standard errors apart the two averages are:

$t = (\text{mean}_1 - \text{mean}_2) \div \text{SE}$, where **$\text{SE} = \sqrt{(s_1^2 \div n_1 + s_2^2 \div n_2)}$**.

Check it on the Business results:

Step	Calculation
Difference in means	\$6,575.99 − \$9,099.14 = −\$2,523.15
Scenario A variance ÷ n	$1{,}651.28^2 \div 150 = 18{,}178$
Scenario B variance ÷ n	$7{,}141.96^2 \div 150 = 340{,}051$
SE	$\sqrt{(18{,}178 + 340{,}051)} = \$598.52$
t	**−2,523.15 ÷ 598.52 = −4.216**

This matches the t your notebook reports for Business (−4.2156). The large gap in averages is divided by a fairly large SE, because Scenario B is so variable. Try the same arithmetic for your own track before trusting the software.

What a p-value is. Picture all the t-values you would see if there were truly no difference. They form a bell-shaped curve centered on zero, and the p-value is the area beyond your observed t in both tails.

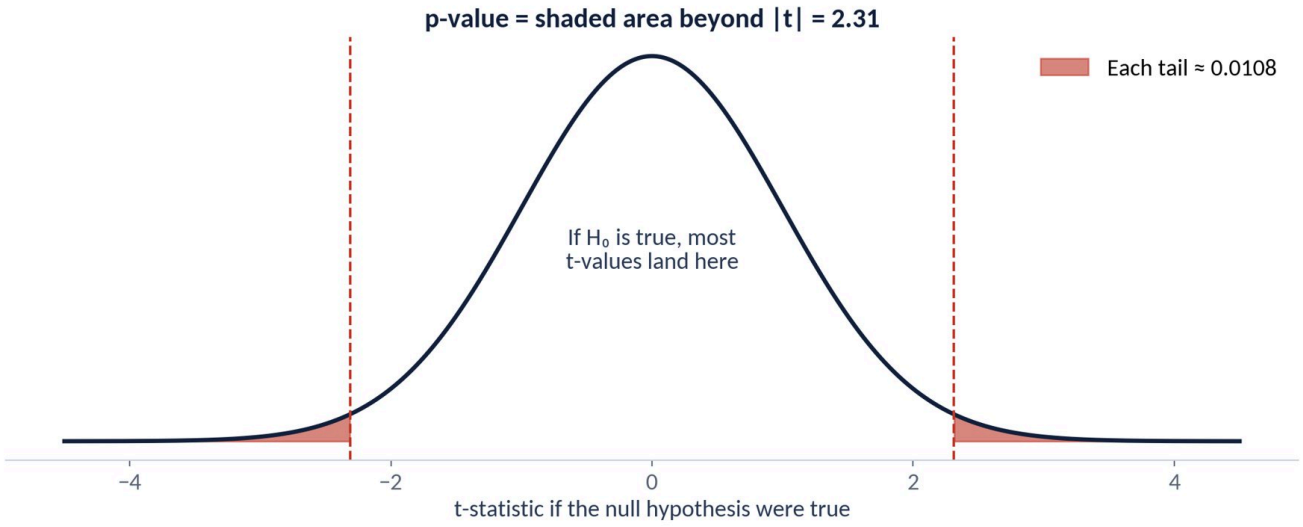


Figure 6.2. For t = 2.31 with 298 degrees of freedom, each tail holds about 1.1%, so p ≈ 0.022.

- **It is** the chance of a gap at least this large if there were truly no difference.
- **It is not** the probability that H_0 is true, or the chance your result is wrong.
- **Type I error:** calling a difference real when it is not. $\alpha = 0.05$ caps this risk at 5%.

- **Type II error:** missing a real difference, more likely with small or noisy samples.

Why 0.05? It is a convention, not a law: accepting a 1-in-20 risk of a false alarm. A decision where false alarms are costly can demand 0.01.

```
t_stat, p_value = stats.ttest_ind(data_1, data_2)   # Group 1: High Demand, Group 2: Low Demand
```

The t-statistic is 136.11. The notebook prints the p-value as 0.000000, but a p-value never equals exactly zero; report it as **p < 0.001**.

Model interpretation. Because p < 0.001, I reject the null hypothesis. Average profit differs significantly between high- and low-demand periods, so demand is the variable worth forecasting and managing.

Caution: significant is not the same as important. With 5,000 periods per group, even a tiny gap would be significant. Always ask how big the difference is in dollars.

6.6 95% confidence intervals

95% CI = $\bar{x} \pm t^* \times s/\sqrt{n}$, where $\bar{x}$ is the sample mean, s the standard deviation, n the sample size, and t* the t critical value.

A 95% confidence interval is the likely range for the **true average**, not the range of individual outcomes.

Group	Lower bound	Upper bound	Width
High demand	$4,773.51	$4,792.21	$18.70
Low demand	$2,099.42	$2,173.33	$73.91

Model interpretation. Both intervals sit well above break-even and do not overlap, so on average both groups are profitable and clearly different. They are narrow because each group has 5,000 periods. Your assignment has 150 simulations per group, so your intervals will be far wider: width reflects both the spread of the data and the sample size.

Where the interval comes from. The margin of error is t* × SE, where t* is the cut-off that leaves 2.5% in each tail of the t-distribution. It is close to 2, and closer as n grows: 2.262 for n = 10, 2.045 for n = 30, 1.976 for n = 150, and 1.960 for n = 5,000.

Step (Business, Scenario A)	Calculation
Standard error	1,651.28 ÷ √150 = $134.83
Margin of error	1.976 × 134.83 = $266.42
Interval	$6,575.99 ± $266.42
Result	**$6,309.57 to $6,842.41**

Why width shrinks with √n. The margin divides by √n, so quadrupling the sample halves the interval. That is why the practice intervals (n = 5,000) are about six times narrower than yours (n = 150) at the same spread.

6.7 Reading intervals: six situations

What you see	What it means
Entirely above $0	The average is likely positive.
Crosses $0	The average could be positive or negative; be cautious.
Entirely below $0	The average is likely negative; a warning sign.
Narrow	A precise, more predictable average.
Wide	An imprecise average; weigh your risk tolerance.
Two intervals overlap	Rely on the t-test before claiming a difference.
Two intervals do not overlap	Strong visual evidence of a real difference.

6.8 Correlation: what drives profit?

Correlation by hand. Five programs, with admin overhead x and net impact y in $ thousands. Both means are 6; dx and dy are each value's distance from its mean.

x	y	dx	dy	dx × dy
2	9	−4	+3	−12
4	8	−2	+2	−4

x	y	dx	dy	dx × dy
6	6	0	0	0
8	5	+2	−1	−2
10	2	+4	−4	−16
Σ		**Σdx² = 40**	**Σdy² = 30**	**−34**

r = Σ(dx × dy) ÷ √(Σdx² × Σdy²) = −34 ÷ √(40 × 30) = −0.98. Correlation measures whether points rise or fall together along a straight line, from −1 to +1. Here, programs with more overhead almost always show less impact.

Correlation is not causation. Overhead may not cause low impact: struggling programs may hire more administrators, or a third factor such as size or location may drive both. Say what a correlation shows, and be careful about what it does not.

In the practice data:

Pair	Correlation	Reading
Demand and profit	+0.93	Strong positive: when demand rises, profit rises.
Holding cost and profit	−1.00	Perfect negative, in both groups.

A correlation of −1.00 is a warning, not a discovery. Holding cost is computed from the same unsold units that reduce sales, so the two must move together. Perfect correlations usually reveal a formula built into the model.

Model interpretation. Profit is driven by demand, and holding cost is the channel through which low demand hurts. Cutting the holding cost, or arranging returns for unsold stock, would reduce the downside directly.

6.9 Reading the histogram and box plot

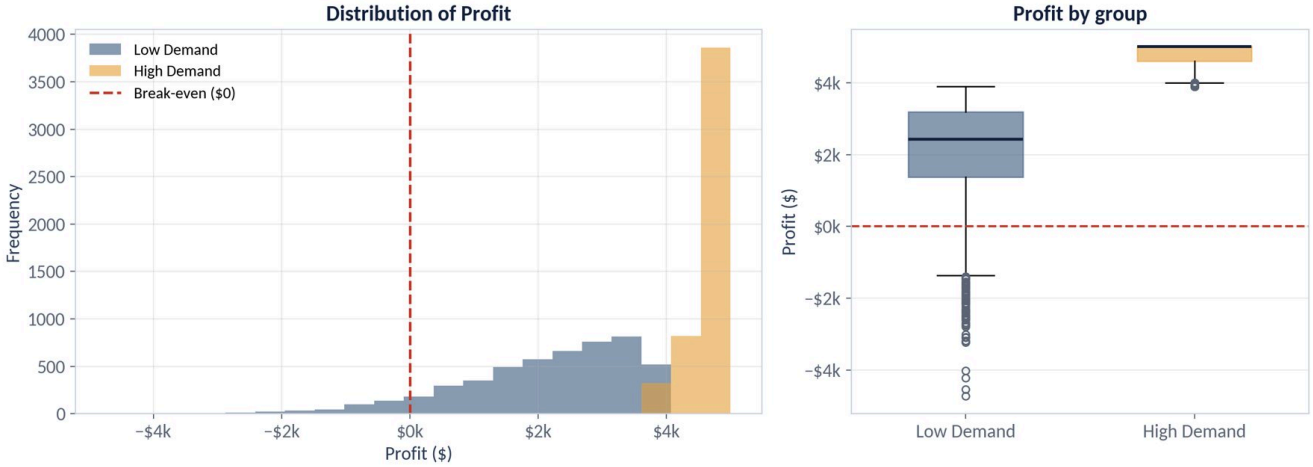


Figure 6.3. Profit by demand group. Read every chart for shape, center, spread, outliers, and how much falls below break-even.

- **Shape:** both groups are left-skewed; high demand piles up at the $5,000 cap.
- **Center:** the median is $5,000 for high demand and $2,420 for low demand.
- **Spread:** the low-demand box and whiskers are far wider.
- **Below $0:** only the low-demand group crosses break-even.

6.10 Putting it together: a model recommendation

The template. *Option ___ has the stronger average outcome, but Option ___ has lower risk. Because the p-value is ___ and the probability of loss / overrun / inefficiency is ___, I recommend ___ for ___ decision-makers.*

Model recommendation (practice data). Ordering 500 units earns an average of $3,460 per period, with a 3.95% chance of a loss. The risk sits entirely in low-demand periods, which average $2,136 and lose money 7.9% of the time; the gap between demand groups is statistically significant ($t = 136.11$, $p < 0.001$). Because holding cost moves one-for-one with lost profit, I recommend keeping the 500-unit order while negotiating a lower holding cost or return terms for unsold stock. This suits managers who can accept roughly a one-in-25 chance of a small loss.

6.11 Manager's Lens: statistical evidence

> **The decision it supports.** Choosing between two options, policies, or operating models using simulated or observed results.
>
> **What it tells you to do.** Weigh four things together: the average, the chance of a bad outcome, whether the difference is significant, and how wide the uncertainty is.
>
> **What a careless reading gets wrong.** Choosing the higher average alone, or calling a difference real when $p \geqslant 0.05$.

- **Business:** an investor needs both expected profit and the odds of losing money before committing capital.
- **Healthcare:** a CFO budgets to the overrun probability, not only to the average margin.
- **Public administration:** a council member must defend both the average impact and the risk of wasting taxpayer money.

Check your understanding 6.1

An analyst compares two options with 150 simulations each. Option X: mean \$12,000, 95% CI −\$1,500 to \$25,500. Option Y: mean \$10,500, 95% CI \$8,000 to \$13,000. Two-sample t-test: $p = 0.18$.

1. Is the difference in averages statistically significant?
2. What does Option X's confidence interval tell you?
3. Which option suits a risk-averse decision-maker, and why?

Answers. (1) **No.** $p = 0.18 \geqslant 0.05$, so we fail to reject H_0; X's higher mean could be chance. (2) X's interval **crosses \$0**, so its true average could be negative, and the wide interval signals low precision. (3) **Option Y**: a similar average, a narrow interval entirely above break-even, and no significant evidence that X is better. A higher average without significance or precision is not a reason to take on more risk.

Chapter 6 summary

- Read every simulation with six questions: mean, spread, probability of a bad outcome, significance, interval, and drivers.
- $p < 0.05$ means a statistically significant difference, which is not the same as a large one; a p-value is never exactly zero.

- A confidence interval is the likely range for the true average; check whether it crosses break-even and how wide it is.

Chapter 7. Your Assignment, Track by Track

This chapter lists every result your notebook should produce, so you can check your code. The interpretations, the comparisons, and the Task 7 recommendation are yours to write: each section ends with the questions you must answer, not the answers.

7.1 One method, three datasets

Track	File	Group column	Outcome	Cost driver	Bad outcome
Business	`investment_simulation.csv`	`Scenario`	`Profit`	`Holding_Cost`	Loss
Healthcare	`hospital_staffing_simulation.csv`	`Staffing_Model`	`Net_Margin`	`Overtime_Cost`	Budget overrun
Public administration	`community_program_simulation.csv`	`Funding_Model`	`Net_Social_Impact`	`Admin_Overhead_Cost`	Inefficiency

Each dataset has 150 simulations per group. Read all three sections: the same method tells a different story each time.

7.2 Business: conservative or aggressive strategy?

Result	Scenario A (Conservative)	Scenario B (Aggressive)
Mean profit (Task 1)	$6,575.99	$9,099.14
Standard deviation	$1,651.28	$7,141.96
Losses (Task 2)	0 of 150 (0.0%)	11 of 150 (7.3%)
95% CI for mean profit (Task 5)	$6,309.57 to $6,842.41	$7,946.85 to $10,251.43
CI width	$532.84	$2,304.58
Correlation: holding cost vs. profit (Task 4)	−0.140	−0.037
Median profit	$6,631	$9,668
Q1 to Q3	$5,378 to $7,723	$4,074 to $14,544

Result	Scenario A (Conservative)	Scenario B (Aggressive)
Minimum / maximum	$1,841 / $10,667	−$8,984 / $28,059
Average ROI	77.8%	72.6%
Average risk score (1–10)	3.2	8.0

Overall mean profit across all 300 simulations: **$7,837.56**. Two-sample t-test (Task 3): **$t = -4.2156$, $p = 3.31 \times 10^{-5}$** (report as $p < 0.001$).

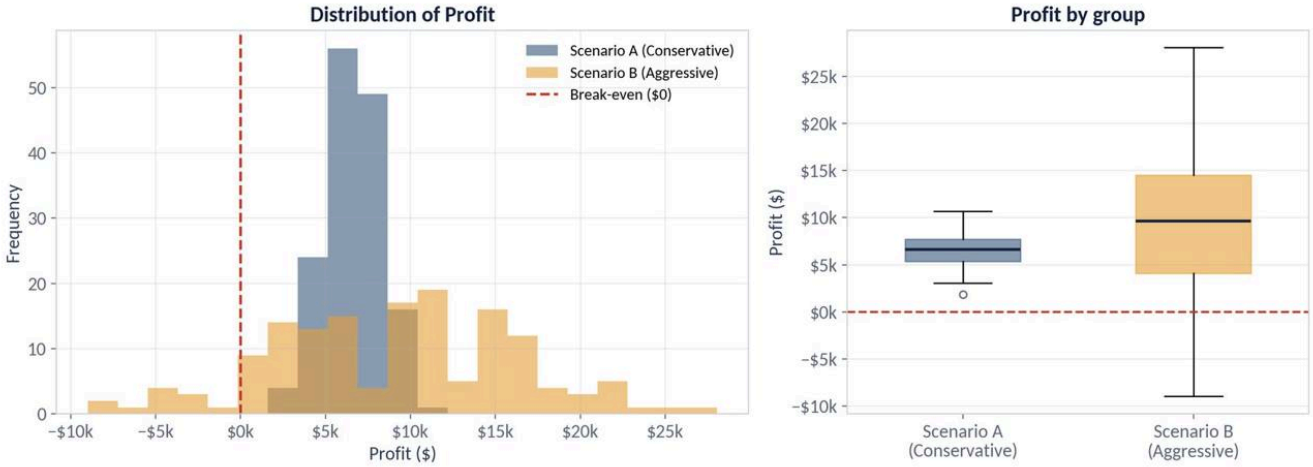


Figure 7.1. Task 6 output for the Business dataset: overlapping histograms and side-by-side box plots of profit.

Your turn. Your decision-maker is an investor who wants return but must survive the bad years.

1. Which scenario has the higher average profit, and by how much? Which is riskier? Connect your answer to the risk-return tradeoff.
2. Do you reject or fail to reject the null hypothesis? What does that mean for an investor?
3. Which confidence interval is wider, do they overlap, and what are the best- and worst-case expected profits?
4. Is each correlation positive or negative, and how strong? What does that mean for cost management?
5. Describe each distribution's shape, center, spread, and outliers. Which scenario would a risk-averse investor prefer?

7.3 Healthcare: which staffing model?

Result	Model A (Standard)	Model B (Enhanced)
Mean net margin (Task 1)	$10,742.27	$13,943.39
Standard deviation	$13,006.25	$10,936.57
Budget overruns (Task 2)	33 of 150 (22.0%)	15 of 150 (10.0%)
95% CI for mean net margin (Task 5)	$8,643.83 to $12,840.70	$12,178.87 to $15,707.90
CI width	$4,196.88	$3,529.03
Correlation: overtime cost vs. net margin (Task 4)	−0.398	−0.108
Median net margin	$10,986	$14,735
Q1 to Q3	$1,427 to $19,493	$5,470 to $22,230
Minimum / maximum	−$17,432 / $52,863	−$15,716 / $41,514
Patient satisfaction (1–100)	70.8	85.5
Staff burnout index (1–10)	7.25	4.48

Overall mean net margin across all 300 days: **$12,342.83**. Two-sample t-test (Task 3): **t = −2.3071, p = 0.0217**.

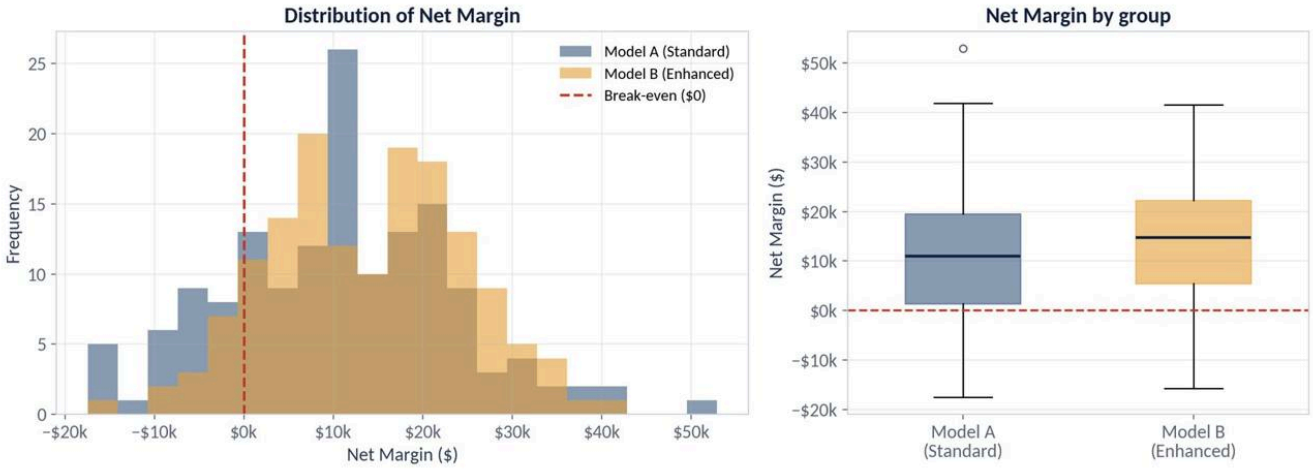


Figure 7.2. Task 6 output for the Healthcare dataset: overlapping histograms and side-by-side box plots of net margin.

Your turn. Your decision-maker is a hospital CFO, accountable for the budget and for patient care.

1. Which model has the higher average margin, and which is more variable? What does variability mean for budgeting?
2. How many times more likely is an overrun in the riskier model? Why would a CFO care even when average margins are positive?
3. Do you reject the null hypothesis at $\alpha = 0.05$? The intervals overlap slightly, yet the t-test reaches a verdict: explain how both can be true.
4. Which model is more sensitive to overtime, and what does that suggest about controlling it?
5. Which model would a risk-averse CFO prefer, weighing the financial evidence alongside patient satisfaction and burnout?

7.4 Public administration: which funding model?

Result	Model A (Standard)	Model B (Enhanced)
Mean economic benefit (Task 1)	$155,671.80	$279,260.76
Standard deviation of economic benefit	$41,907.11	$53,141.76
Inefficiency scenarios (Task 2)	41 of 150 (27.3%)	16 of 150 (10.7%)
Mean net social impact	$23,096.41	$48,290.97
95% CI for mean net social impact (Task 5)	$16,656.78 to $29,536.04	$41,546.17 to $55,035.77
CI width	$12,879.26	$13,489.60
Correlation: admin overhead vs. net social impact (Task 4)	−0.626	−0.265
Average overhead %	23.4%	12.5%
Median net social impact	$20,314	$46,972
Q1 to Q3	−$3,473 to $49,322	$19,282 to $74,176
Minimum / maximum	−$81,851 / $152,461	−$72,322 / $150,316
Participants served	121	260

Result	Model A (Standard)	Model B (Enhanced)
Community satisfaction (1–100)	69.9	80.9
Average SROI ratio	1.19	1.21

Overall mean economic benefit across all 300 simulations: **$217,466.28**. Two-sample t-test on net social impact (Task 3): **$t = -5.3387$, $p = 1.86 \times 10^{-7}$** (report as $p < 0.001$).

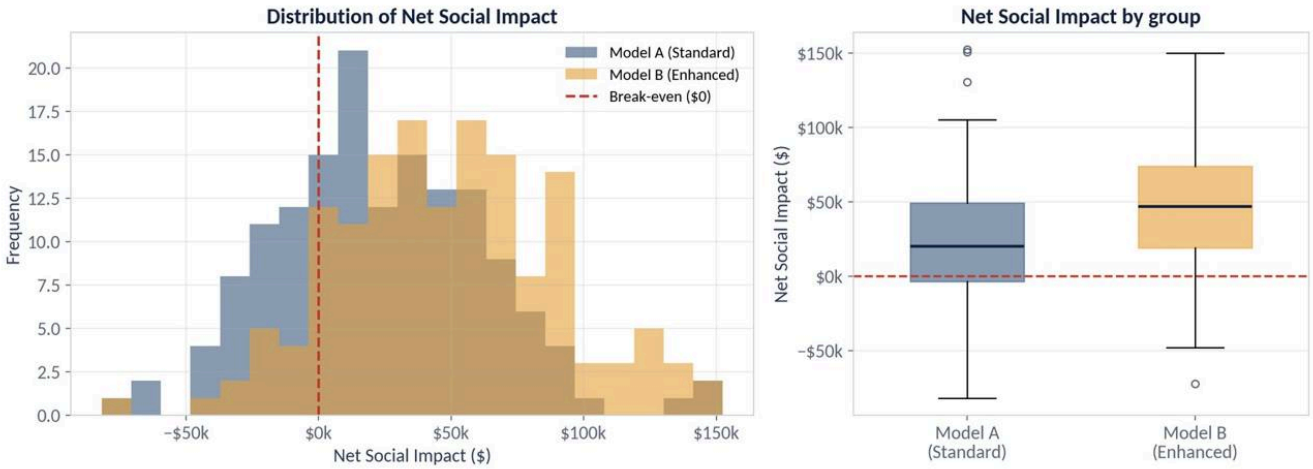


Figure 7.3. Task 6 output for the Public Administration dataset: overlapping histograms and side-by-side box plots of net social impact.

Your turn. Your decision-maker is the City Council, which spends taxpayer money in public view.

1. Which model generates greater benefit, how much more on average, and which is more predictable?
2. How many times more likely is an inefficiency in the riskier model? Why should council members care even if average impact is positive?
3. Do you reject the null hypothesis? Which interval is wider, do they overlap, and does either include break-even?
4. Which model is more efficient, and how does overhead relate to net social impact in each?
5. Compare the SROI ratios with the other evidence. What would a reader who looked only at SROI miss?

Chapter 7 note

If your numbers differ from these, check the file name, the group column, and that you filtered each group with the exact labels (`Scenario_A`, `Staffing_Model_A`, `Funding_Model_A`).

Chapter 8. From Statistics to a Recommendation

Numbers persuade only when they answer the decision-maker's question. This chapter shows how to turn your results into a recommendation someone can act on.

8.1 Structure your Task 7 recommendation

1. **Recommendation.** State the option you recommend, in one sentence, for a named decision-maker.
2. **Statistical evidence.** Cite at least four statistics: means, the probability of a bad outcome, the p-value, the confidence intervals.
3. **Practical impact.** Investors: return and risk. Healthcare: patients and staff. Public administration: residents served and their satisfaction.
4. **Accountability.** Address cost control: holding costs, overtime, or administrative overhead.
5. **Limitations.** Name your assumptions and the extra information that would improve the decision.

Public Administration's Task 7 requires all five parts by name. Business and Healthcare should cover the same ground.

8.2 Weak versus strong recommendations

Both examples use the practice data. Your recommendation must use your own track's results.

	Weak	Strong
Example	"Ordering 500 units is better because it makes more money on average. The p-value is 0.000000, so it is definitely right."	"Keep the 500-unit order: it averages $3,460 with a 3.95% chance of loss. Losses come only from low-demand periods ($p < 0.001$), so reduce holding costs to protect the downside."

	Weak	Strong
Evidence	Only the average	The average and the risk together
p-value	Misread as zero and as proof	Read correctly and cautiously
Action	No risk, no decision-maker, no action	Names the driver and an action

8.3 Two fixes for cleaner notebook output

Report tiny p-values correctly. The template's `:.6f` format rounds a tiny p-value to 0.000000. Use scientific notation, and write "p < 0.001" in your interpretation.

```
print(f"P-value: {p_value:.2e}")
```

Keep the box plot beside the histogram. `df.boxplot(..., by=...)` opens a new figure on its own, leaving an empty panel next to your histogram. Tell it to draw on the current axes.

```
df.boxplot(column=METRIC_COL, by=GROUP_COL, ax=plt.gca())
```

A report that says "p = 0" or shows an empty chart loses credibility before the argument begins. Clean output is part of the evidence.

Check your understanding 8.1

A county compares two job-training programs, 150 simulations each. Program R: mean net impact $40,000, inefficiency 18%, 95% CI $30,000 to $50,000. Program S: mean net impact $42,000, inefficiency 5%, 95% CI $38,000 to $46,000. Two-sample t-test: p = 0.61.

1. Can the county claim Program S has the higher average impact?
2. Which program is lower-risk, and what evidence shows it?
3. Write a one-sentence recommendation using the template.

Answers. (1) **No.** With p = 0.61, the $2,000 gap is not statistically significant. (2) **Program S**: inefficiency of 5% versus 18%, and a narrower interval ($8,000 wide versus $20,000). (3) One strong answer: *"The programs' averages are not significantly different (p = 0.61), but Program S has lower risk, with a 5% inefficiency rate versus 18% and a narrower interval, so I recommend Program S for a county that must protect taxpayer funds."* The recommendation rests on risk, not on a small, non-significant difference in averages.

8.4 Before you submit: six checks

- [ ] Is one option's average higher, and by how much?
- [ ] Is the probability of a bad outcome higher or lower?
- [ ] Did the t-test show a significant difference ($p < 0.05$)?
- [ ] Are the confidence intervals narrow or wide?
- [ ] Does any interval cross break-even?
- [ ] Do the charts show major outliers or a large spread?

A strong answer explains both the numbers and what they mean for the decision-maker.

Chapter 8 summary

- A recommendation names an option, a decision-maker, the evidence, the impact, the cost controls, and the limits.
- Risk and significance belong in every argument, alongside the average.
- Every tool in this guide serves one purpose: helping a manager choose well when the future is uncertain.

Appendix A. Key Formulas

Complement and "at least once" (Chapter 1)

$P(\text{not } A) = 1 - P(A)$ · $P(\text{at least once in n periods}) = 1 - (1 - p)^n$

Multiplication rule (Chapter 1)

$P(A \text{ and } B) = P(A) \times P(B \mid A)$

Expected monetary value and risk premium (Chapter 2)

$EMV = \Sigma\, p_i \times x_i$ · Risk premium = EMV − CE

Value of information (Chapter 3)

EVPI = EV with perfect information − EMV · EVSI = EV with sample information − EMV · Efficiency = EVSI ÷ EVPI

Bayes' rule (Chapter 3)

$P(A \mid B) = P(A \text{ and } B) \div P(B)$

Expected utility (Chapter 4)

$EU = \Sigma\, p_i \times U(x_i)$

95% confidence interval for a mean (Chapter 6)

$\bar{x} \pm t^* \times s/\sqrt{n}$, with t^* taken from the t-distribution at 0.975 and $n - 1$ degrees of freedom

Addition rule (Chapter 1)

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ · Independent events: $P(A \text{ and } B) = P(A) \times P(B)$

Variance and standard deviation of a decision (Chapter 2)

$\text{Variance} = \Sigma\, p_i \times (x_i - \text{EMV})^2$ · $\text{SD} = \sqrt{\text{variance}}$

Sample standard deviation (Chapter 6)

$s = \sqrt{[\ \Sigma\, (x_i - \bar{x})^2 \div (n - 1)\ ]}$

z-score (Chapter 5)

$z = (x - \mu) \div \sigma$

Standard error and two-sample t-statistic (Chapter 6)

SE of a mean $= s \div \sqrt{n}$ · $t = (\text{mean}_1 - \text{mean}_2) \div \sqrt{(s_1^2 \div n_1 + s_2^2 \div n_2)}$

Correlation (Chapter 6)

$r = \Sigma(dx \times dy) \div \sqrt{(\Sigma dx^2 \times \Sigma dy^2)}$, where dx and dy are each value's distance from its mean

Appendix B. Glossary

Term	Meaning
Addition rule	P(A or B) = P(A) + P(B) − P(A and B); subtract the overlap so it is not counted twice.
Bayes' rule	A way to revise a probability after new evidence arrives.
Break-even	The point where an outcome is exactly $0; below it is a loss, overrun, or inefficiency.
Break-even probability	The probability at which two options have the same EMV; the decision flips on either side of it.
Certainty equivalent (CE)	The guaranteed amount that feels exactly as good as a gamble.

Term	Meaning
Chance node	A circle in a decision tree marking an uncertain event.
Conditional probability	The probability of an event given that another has happened, written P(B | A).
Confidence interval (95%)	The likely range for the true average, not for individual outcomes.
Correlation	A number from −1 to +1 showing whether two variables move together along a straight line.
Decision node	A square in a decision tree marking a choice the manager controls.
Decision tree	A diagram of decisions, uncertain events, and payoffs, solved by folding back.
Expected monetary value (EMV)	The probability-weighted average payoff of an option.
Expected utility (EU)	The probability-weighted average of an option's utilities.
EVPI	The expected value of perfect information; the most any information can be worth.
EVSI	The expected value of sample (imperfect) information.
Independent events	Events where one tells you nothing about the other.
Law of large numbers	As the number of trials grows, an average or percentage settles on its true value.
Lottery (standard gamble) method	Finding utilities by asking at what winning probability a manager is indifferent between a gamble and a sure amount.
Margin of error	t* × standard error; half the width of a confidence interval.
Monte Carlo simulation	Running a model thousands of times with random inputs to see the full range of outcomes.

Term	Meaning
Mutually exclusive events	Events that cannot happen together.
Normal distribution	The bell curve, defined by its mean μ and standard deviation σ.
Null hypothesis (H_0)	The assumption that two groups have the same true average.
p-value	If H_0 were true, the chance of seeing a difference at least as large as the one observed.
Regret	What the best choice for an outcome would have earned, minus what you actually earned.
Risk premium	EMV minus the certainty equivalent: the expected value given up for certainty.
Risk-averse / neutral / seeking	CE below / equal to / above the EMV.
Sampling distribution	The distribution of a statistic, such as a mean, across many samples.
SROI	Social return on investment: benefit divided by cost.
Standard deviation	A measure of spread; higher means less predictable outcomes.
Standard error (SE)	The standard deviation of a sample mean: $s \div \sqrt{n}$.
Statistical significance	A difference unlikely to be due to chance, usually $p < 0.05$.
t-test (two-sample)	A test of whether two group averages differ.
Type I / Type II error	Calling a difference real when it is not / missing a difference that is real.
Utility	A score for how good or bad an outcome is for a particular organization.
Variance	The average squared distance from the mean; its square root is the standard deviation.
z-score	How many standard deviations a value sits from the mean: $(x - \mu) \div \sigma$.